

1. Given a string *s*, a partitioning of the string is a palindrome partitioning if every sub-string of the partition is a palindrome. Determine the fewest cuts needed for palindrome partitioning of the given string.

Examples:

Input: *s* = "geek"

Output: 2

Explanation: We need to make minimum 2 cuts, i.e., "g | ee | k".

Input: *s* = "aaaa"

Output: 0

Input: *s* = "ababbbabbababa"

Output: 3

Explanation: We need to make minimum 3 cuts, i.e., "aba | bb | babbab | aba".

2. Given a square grid of size *N*, each cell of which contains an integer cost that represents a cost to traverse through that cell, we need to find a path from the top left cell to the bottom right cell by which the total cost incurred is minimum. From the cell (*i*,*j*) we can go (*i*,*j*-1), (*i*, *j*+1), (*i*-1, *j*), (*i*+1, *j*).

Examples :

Input: grid = [[9,4,9,9],
 [6,7,6,4],
 [8,3,3,7],
 [7,4,9,10]]

Output: 43

Explanation: The minimum cost is - 9 + 4 + 7 + 3 + 3 + 7 + 10 = 43.

Input: grid = [[4,4],
 [3,7]]

Output: 14

3. Given two distinct words *startWord* and *targetWord*, and a list denoting *wordList* of unique words of equal lengths. Find the length of the shortest transformation sequence from *startWord* to *targetWord*. Keep the following conditions in mind:
A word can only consist of lowercase characters.
Only one letter can be changed in each transformation.
Each transformed word must exist in the *wordList* including the *targetWord*.
startWord may or may not be part of the *wordList*

Example 1:

Input:

wordList = {"des","der","dfr","dgt","dfs"}

startWord = "der", *targetWord* = "dfs",

Output:

3

Explanation:

The length of the smallest transformation sequence from "der" to "dfs" is 3
i.e "der" -> "dfr" -> "dfs".

Example 2:

Input:

wordList = {"geek", "gefk"}

startWord = "gedk", *targetWord* = "geek",

Output:

2

Example 3:

Input:

wordList = {"poon", "plee", "same", "poie", "plea", "plie", "poin"}

startWord = "toon", targetWord = "plea",

Output: 7

4. Consider the generalized Fibonacci number g , which is dependent on a , b and c as follows :-
 $g(1) = 1$, $g(2) = 1$. For any other number n , $g(n) = a \cdot g(n-1) + b \cdot g(n-2) + c$. For a given value of m , determine $g(n) \% m$.

Example 1:

Input:

$a = 3$

$b = 3$

$c = 3$

$n = 3$

$m = 5$

Output:

4

Explanation:

$g(1) = 1$ and $g(2) = 1$

$g(3) = 3 \cdot g(2) + 3 \cdot g(1) + 3 = 3 \cdot 1 + 3 \cdot 1 + 3 = 9$

We need to return answer modulo 5, so $9 \% 5 = 4$, is the answer.

Example 2:

Input:

$a = 2$

$b = 2$

$c = 2$

$n = 4$

$m = 100$

Output:

16

Explanation:

$g(1) = 1$ and $g(2) = 1$

$g(3) = 2 \cdot g(2) + 2 \cdot g(1) + 2 = 2 \cdot 1 + 2 \cdot 1 + 2 = 6$

$g(4) = 2 \cdot g(3) + 2 \cdot g(2) + 2 = 2 \cdot 6 + 2 \cdot 1 + 2 = 16$

We need to return answer modulo 100, so $16 \% 100 = 16$, is the answer.

5. Given an array `arr[]` containing only 0s, 1s, and 2s. Sort the array in ascending order.

Examples:

Input: `arr[] = [0, 1, 2, 0, 1, 2]`

Output: `[0, 0, 1, 1, 2, 2]`

Explanation: 0s, 1s and 2s are segregated into ascending order.

Input: `arr[] = [0, 1, 1, 0, 1, 2, 1, 2, 0, 0, 0, 1]`

Output: `[0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 2, 2]`